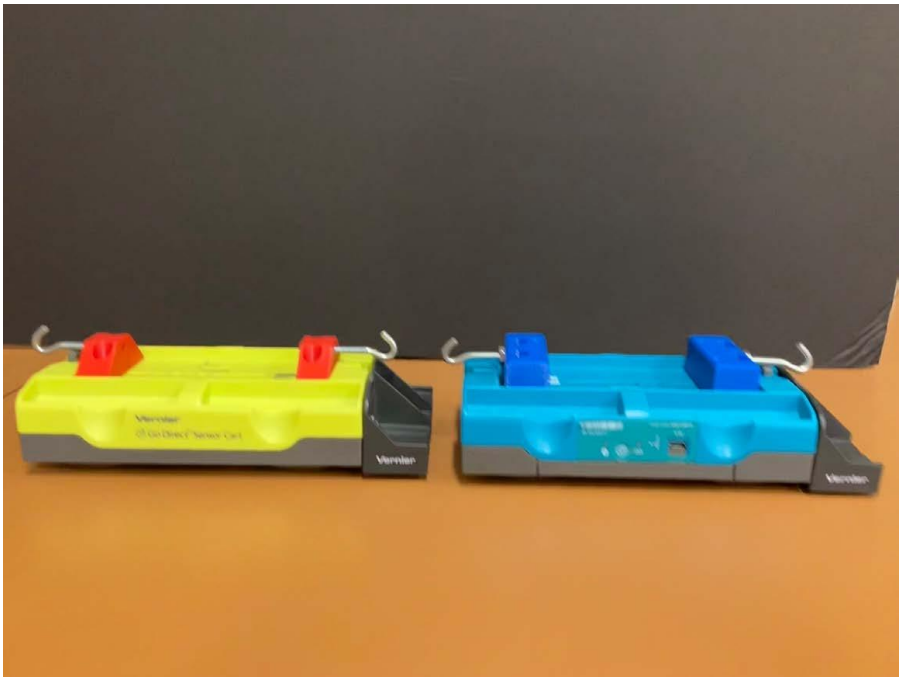
Lab 4: Free Response of a 1-DOF Spring-Mass-Damper System

Fall 2026 | Laboratory week: October 26-30 | Prof. Ehsan Roohi



Why this experiment matters

- **Vehicle suspensions:** ring-down after a bump reveals oscillation frequency and settling.
- **Machine mounts and instruments:** damping limits motion after impacts and protects sensitive components.
- **Structural and aerospace testing:** a measured decay identifies a dominant mode and can reveal changes in stiffness or attachment condition.

Lab 4 is a controlled ring-down system-identification experiment. Students infer frequency and damping from a measured position record and test whether changes in mass, stiffness, and magnetic damping follow the 1-DOF model.

Learning outcomes

- Measure the moving mass and individual spring stiffnesses.
- Predict and measure damped and undamped natural frequency.
- Estimate decay rate, damping ratio, and equivalent damping coefficient from same-sign peaks.
- Use an FFT as an independent frequency check and explain its limits.
- Preserve analysis-ready evidence for the combined Labs 4-5 report.

1. Equipment, safety, and physical model

Equipment: Vernier dynamics track and Go Direct Sensor Cart; Vernier Force Sensor; four springs; hex-bar masses; eddy-current magnetic damper pieces; Graphical Analysis; balance; ruler or caliper; computer with MATLAB.

Safety and care

- Keep the cart on the track and keep hands clear of the end stops during release.
- Do not overstretch a spring. For the horizontal setup, each spring should remain approximately 15 cm long near equilibrium.
- Support hanging masses securely during the spring test. Remove masses before rearranging the setup.
- Release the cart without pushing it. Catch the cart before any hard collision.
- Do not place ferromagnetic tools near the magnetic damper gap.

Model and definitions

Measure cart displacement $x(t)$ from equilibrium. The total moving mass m includes the cart and every attached hex bar and damper component. The left and right springs both oppose a small horizontal displacement, so $k_{eq} = k_L + k_R$. The equivalent viscous damping coefficient is c.

$$m\,\mathrm{d}^2x/\mathrm{d}t^2 + c\,\mathrm{d}x/\mathrm{d}t + k_{eq}x = 0$$

$$\omega_n = \sqrt{(k_{eq}/m)},\quad f_n = \omega_n/(2\pi),\quad \zeta = c/(2\sqrt{(m\,k_{eq})})$$

$$\omega_d = \omega_n\sqrt{(1-\zeta^2)},\quad \alpha = \zeta\omega_n = c/(2m)$$

The magnetic brake is treated as approximately velocity-proportional over the tested range. Rolling and dry friction can make the zero-magnet response depart from a perfect exponential. State this limitation rather than forcing a poor fit.

2. Before collection: identify every case

Create the metadata table before opening Graphical Analysis. Use one row per run. Every file name must match one row.

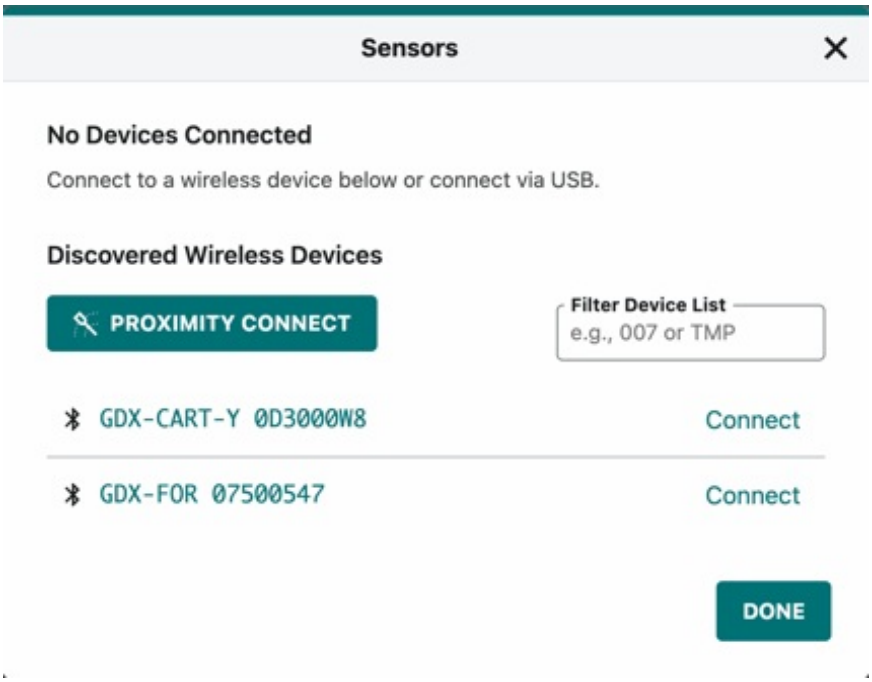
Run	cart + attached mass (kg)	left spring	right spring	dampers	x0 (m)	fs (Hz)	duration (s)	CSV file
1								
2								
3								
4								
5								
6								
7								

Measurement settings

- Pair each sensor by serial number. Do not assume the first device shown belongs to your station.
- For the cart, record the actual sampling rate and check that it gives at least 15-20 samples per cycle. The prior manual used rates no greater than 25 Hz; adequacy depends on the measured frequency.
- Choose a record long enough to show several cycles and most of the decay.
- Use the same small initial displacement for comparisons unless initial displacement is the intended variable.
- Zero the cart at equilibrium before every run and re-zero if the cart moves.

Naming convention

Group_Run##_Mass##_SpringL##_SpringR##_Damper#.csv. Example: *G2_Run04_M1p24_SA_SA_D2.csv*. Do not overwrite or edit raw CSV files.



Graphical Analysis sensor selection. Confirm the cart/force-sensor serial number before collection.

3. Measure the four spring stiffnesses

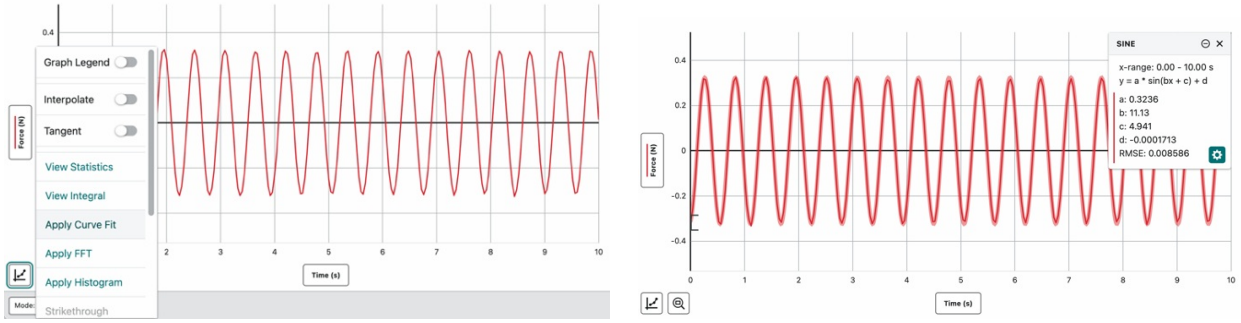
A measured spring value is required because age, manufacturing variation, and attachments can shift the nominal 10.7 N/m value.

Preferred transparent method: static force-extension slope

- Mount the Force Sensor securely and hang one spring vertically.
- Record force and extension for at least five safe loads, including zero load.
- Fit $F = k\Delta x + b$. The slope gives k [N/m]. Report the fit and do not use a single point when several are available.
- Repeat for all four springs and label them physically so the cart configuration can be reconstructed.

Dynamic cross-check: hanging-mass ring-down

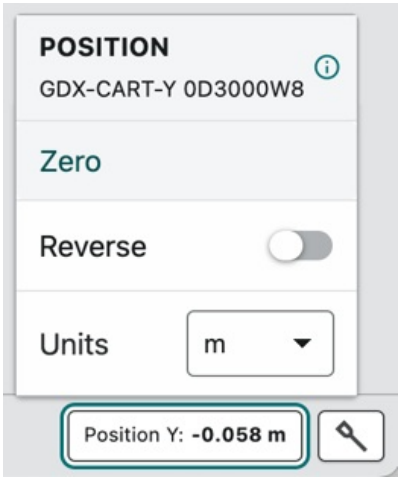
- Hang a known total moving mass m_h, give a small vertical displacement, and record 5-10 s at a rate no greater than 25 Hz.
- Use Apply Curve Fit > sinusoid or export the force-time record. The fitted angular frequency is approximately ω_d.
- For light damping, $k \approx m_h \omega_d^2$. If decay rate α is measured, use $k = m_h(\omega_d^2+\alpha^2)$. Include hanger and attachment mass in m_h.
- If static and dynamic estimates disagree materially, check mass, extension reference, spring motion, and attachment effects.

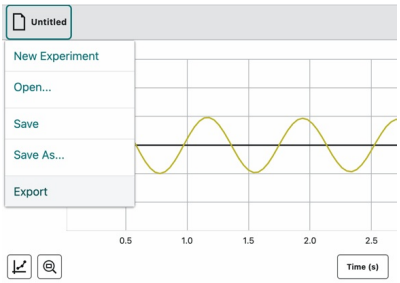


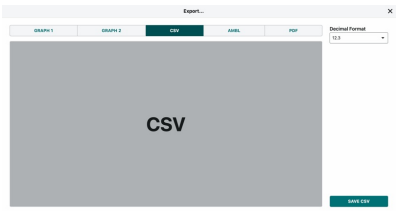
Graphical Analysis curve-fit menu and a representative sinusoidal fit. Record the fitted frequency parameter and units.

4. Record the cart free response

Step	Action	Required detail
1	Level and inspect	Level the track, confirm clear travel, and check that both spring attachments are secure.
2	Build the base case	Place the cart between two labeled springs. Record total moving mass, spring IDs, and zero dampers.
3	Pair and zero	Pair the correct cart by serial number. Use position in meters. Zero at the equilibrium position.
4	Prepare the release	Move the cart by a measured small displacement. Check that no spring is slack or overstretched.
5	Collect and release	Click Collect, then release without a push. Do not release first and do not strike the cart.
6	Record the decay	Continue until the response nearly settles. Stop before the cart reaches an end stop.
7	Export immediately	Export CSV with the planned name. Plot it immediately and verify several clean cycles.
8	Preserve and reset	Do not edit the raw file. Re-zero, update metadata, and continue with the next case.







Zero position, export command, and CSV export. Confirm units and destination before closing the run.

5. Seven-case experimental matrix

Case	Mass	Spring pair	Dampers	Purpose and expected trend
1	base	base	0	reference
2	+ mass	base	0	frequency decreases
3	++ mass	base	0	confirm mass trend
4	base	base	2	decay becomes faster
5	base	base	4	confirm damping trend
6	base	softer pair	0	frequency decreases
7	base	stiffer pair	0	frequency increases

Comparison rules

- Change one factor at a time. Return other factors to the base condition.
- Use the same small initial displacement and release method for comparable cases.
- A mounted damper component adds moving mass. Include this mass when predicting frequency.
- Complete more than seven cases if a repeat is needed or if the GTA requests a diagnostic run.

Acceptance check before leaving

- Several complete cycles are visible and the cart never collides with an end stop.
- The envelope is physically plausible and contains no missing sample block.
- The file opens and plots correctly outside Graphical Analysis.
- Metadata and file name agree.
- **Fully analyze one case and show the time trace, decay fit, FFT, and summary to the GTA before leaving.**

Preserve every raw CSV and the completed metadata table. Lab 4 and Lab 5 share one report, so a missing condition cannot be reconstructed later.

6. Required analysis

Use **MIE402_Lab4_Analyze_Free_Response.m**. The code removes equilibrium offset and small drift, selects same-sign peaks, fits the logarithmic decay, computes the FFT, and writes figures plus a summary CSV.

Output	Method	Report meaning
Damped period and frequency	mean spacing of same-sign peaks	T_d and f_d
Decay rate	slope of ln|peak| versus time	slope = -alpha
Damping ratio	zeta = alpha/omega_n	dimensionless decay strength
Equivalent damping	c = 2m alpha	N s/m
FFT peak	largest displacement-spectrum peak in search band	cross-check near f_d
Theoretical frequency	(1/2pi)sqrt(k/m)	prediction from measured mass and springs

Log-decrement cross-check

For positive peaks separated by N complete cycles, $\delta = (1/N)\ln(|x_i|/|x_{i+N}|)$ and $\zeta = \delta/\sqrt{(4\pi^2+\delta^2)}$. A regression through several log peaks is preferred because it uses more data. Report R^2 and inspect residuals.

FFT limitation

A finite decaying signal produces a broadened spectral peak near f_d. FFT checks frequency but does not replace the decay fit for damping. Keep displacement-spectrum units in meters. Do not multiply displacement or velocity spectra by $180/\pi$.

Required summary table

For every case report: total mass, k_L, k_R, k_{eq}, damper count, sample rate, duration, f_d from peaks, FFT peak, f_n, zeta, alpha, c, log-fit R^2, theoretical f_n, and percent difference.

7. Figures and combined Labs 4-5 report evidence

Each figure must carry a claim. Use readable axes, units, run identity, and an annotation for the extracted value.

Evidence	Minimum content
Spring calibration	force-extension data and fitted slope for four springs; uncertainty or fit quality
Representative time record	x(t), selected same-sign peaks, fitted envelope, run metadata
Log-peak regression	ln|peak| versus time, fitted line, alpha and R^2
Representative spectrum	one-sided displacement FFT, annotated peak frequency
Seven-case table	predicted/measured frequency and damping quantities
Trend comparison	effect of mass, stiffness, and damper count with theory-based explanation

Discussion questions

- Does measured frequency follow the square-root dependence on stiffness and inverse square-root dependence on mass?
- Does adding magnets mainly change decay, or does added moving mass also shift frequency?
- Where do peak-spacing, FFT, and fitted-model estimates disagree, and what measurement or model limitation explains the difference?
- Does the no-magnet response follow an exponential envelope? If not, is rolling/Coulomb friction a plausible cause?
- Which uncertainty dominates the theoretical frequency: moving mass, spring stiffness, or analysis choice?

The report should contain equations that support the analysis and discussion. Avoid unrelated vibration theory. Define every symbol at first use.

8. Troubleshooting and exit checklist

Observation	Likely cause	Action
frequency estimates disagree	too short record, wrong peaks, drift, or multiple motion	inspect x(t), use same-sign peaks, extend record
few peaks found	large damping, threshold too high, or poor release	inspect raw plot; repeat with clean release if possible
nonexponential zero-magnet decay	rolling or dry friction	report model limitation; do not force a high-quality exponential claim
cart drifts to one side	track not level or zero set away from equilibrium	level, re-center, re-zero, repeat
unexpected frequency shift with dampers	damper hardware added mass	include every moving damper component in m
CSV columns not recognized	export format or column names differ	open CSV, identify numeric time/position columns, preserve raw file

Exit checklist

- Four spring stiffness values and labels are saved.
- At least seven accepted cart runs are saved with complete metadata.
- Raw CSV files open and remain unchanged.
- One case has a time trace, peak selection, decay fit, FFT, and summary row.
- The GTA has reviewed that analyzed case.
- The group has two backup copies in separate locations.

Files supplied on the course site

- Pre-Lab 4 Colab notebook: predictions, analytical/numerical comparison, decay fit, and FFT.
- MIE402_Lab4_Analyze_Free_Response.m: analysis of one Vernier CSV.
- This Lab 4 instruction PDF and the Lab 4 theory lecture.